

Name: Charmi Gaddale

Roll Number: 23071A0516

AIM:Implementation of real-world case study - Amazon Food Review (or) Personalized Cancer Diagnosis.

Dataset: Amazon Food Reviews

```
#Upload Dataset
from google.colab import files
uploaded = files.upload()
```

Choose files Reviews.csv

Reviews.csv(text/csv) - 300904694 bytes, last modified: 15/04/2026 - 100% done

Saving Reviews.csv to Reviews.csv

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import re
import nltk

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
nltk.download('stopwords')
from nltk.corpus import stopwords
import re

stop_words = set(stopwords.words('english'))
stop_words.remove('not')
```

[nltk_data] Downloading package stopwords to /root/nltk_data...

[nltk_data] Package stopwords is already up-to-date!

```
data = pd.read_csv("Reviews.csv")
data.head()
```

		Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	HelpfulnessDenominator	Score	Time	Summ
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian		1	1	5	1303862400	G Qui Dog F
1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa		0	0	1	1346976000	No Adverti

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 568454 entries, 0 to 568453
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Id                   568454 non-null  int64
1   ProductId           568454 non-null  object
2   UserId              568454 non-null  object
```

```
3 ProfileName      568428 non-null object
4 HelpfulnessNumerator  568454 non-null int64
5 HelpfulnessDenominator 568454 non-null int64
6 Score           568454 non-null int64
7 Time            568454 non-null int64
8 Summary         568427 non-null object
9 Text            568454 non-null object
```

dtypes: int64(5), object(5)
memory usage: 43.4+ MB

```
data['Score'].value_counts()
```

	count
Score	
5	363122
4	80655
1	52268
3	42640
2	29769

dtype: int64

```
# Convert Scores to Sentiment Labels
# We convert ratings into positive and negative sentiment.
def convert_sentiment(score):
    if score <= 2:
        return 0 # negative
    elif score >= 4:
        return 1 # positive
    else:
        return np.nan
```

```
data['Sentiment'] = data['Score'].apply(convert_sentiment)
```

```
data = data.dropna(subset=['Sentiment'])
```

```
data['Sentiment'].value_counts()
```

	count
Sentiment	
1.0	443777
0.0	82037

dtype: int64

```
# Use Only Required Columns
# We only need the review text and sentiment.
data = data[['Text', 'Sentiment']]
data.head()
```

1 to 5 of 5 entries Filter 📄 ?

index	Text	Sentiment
0	I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it smells better. My Labrador is finicky and she appreciates this product better than most.	1.0
1	Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor intended to represent the product as "Jumbo".	0.0
2	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with powdered sugar. And it is a tiny mouthful of heaven. Not too chewy, and very flavorful. I highly recommend this yummy treat. If you are familiar with the story of C.S. Lewis' "The Lion, The Witch, and The Wardrobe" - this is the treat that seduces Edmund into selling out his Brother and Sisters to the Witch.	1.0
3	If you are looking for the secret ingredient in Robitussin I believe I have found it. I got this in addition to the Root Beer Extract I ordered (which was good) and made some cherry soda. The flavor is very medicinal.	0.0
4	Great taffy at a great price. There was a wide assortment of yummy taffy. Delivery was very quick. If your a taffy lover, this is a deal.	1.0

#Text Preprocessing

Text must be cleaned before training models.

Steps:

convert to lowercase

remove punctuation

remove stopwords

from nltk.corpus import stopwords

stop_words = set(stopwords.words('english'))

Function to clean review text

def clean_text(text):

convert text to lowercase

text = text.lower()

remove numbers and punctuation

text = re.sub(r'^a-zA-Z', ' ', text)

split text into words

words = text.split()

remove stopwords

words = [word for word in words if word not in stop_words]

join words back into sentence

return " ".join(words)

data['Clean_Text'] = data['Text'].apply(clean_text)

data[['Text', 'Clean_Text', 'Sentiment']].head()

	Text	Clean_Text	Sentiment	
0	I have bought several of the Vitality canned d...	bought several vitality canned dog food produc...	1.0	
1	Product arrived labeled as Jumbo Salted Peanut...	product arrived labeled jumbo salted peanuts p...	0.0	
2	This is a confection that has been around a fe...	confection around centuries light pillowy citr...	1.0	
3	If you are looking for the secret ingredient i...	looking secret ingredient robitussin believe f...	0.0	
4	Great taffy at a great price. There was a wid...	great taffy great price wide assortment yummy ...	1.0	

#vectorization

from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(max_features=5000, ngram_range=(1,2))

X = vectorizer.fit_transform(data['Clean_Text'])

y = data['Sentiment']

#Train Test Split

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
 X, y, test_size=0.2, random_state=42
)

```
#Train Logistic Regression
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)
```

```
▼ LogisticRegression ⓘ ?
LogisticRegression(max_iter=1000)
```

```
#Predict
y_pred = model.predict(X_test)
```

```
#Evaluate Model
from sklearn.metrics import accuracy_score, classification_report

print("Accuracy:", accuracy_score(y_test, y_pred))

print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.9315633825585045
      precision    recall  f1-score   support

      0.0         0.85      0.68      0.76      16379
      1.0         0.94      0.98      0.96      88784

 accuracy          0.93      0.93      0.93      105163
 macro avg         0.90      0.83      0.86      105163
 weighted avg         0.93      0.93      0.93      105163
```

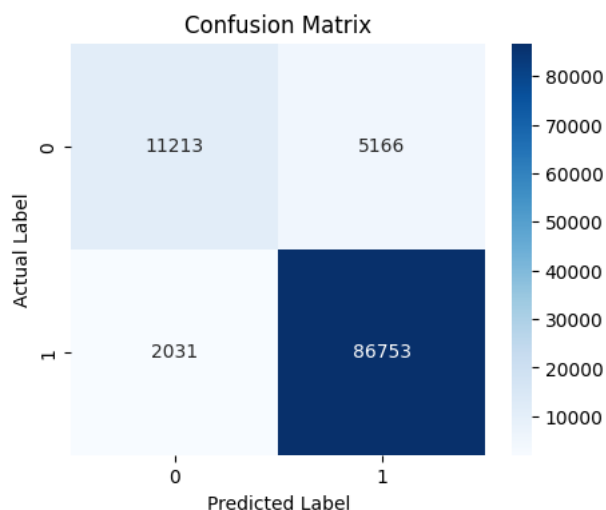
```
#Confusion Matrix
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')

plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.title("Confusion Matrix")

plt.show()
```



```
#Test Model with a New Review
```

```
review = "This product tastes amazing but I got oxidized product because of this my bo"
```

```

review = [ "this product tastes amazing but i got expired product because of this my head
clean = clean_text(review[0])

vector = vectorizer.transform([clean])

prediction = model.predict(vector)

if prediction[0] == 1:
    print("Sentiment: Positive Review")
else:
    print("Sentiment: Negative Review")

```

Sentiment: Negative Review

```

#Test Multiple Reviews
reviews = [
    "This food is delicious and my dog loves it",
    "Worst product ever, completely disappointed"
]

for r in reviews:

    clean = clean_text(r)
    vector = vectorizer.transform([clean])

    pred = model.predict(vector)

    if pred[0] == 1:
        print(r, "-> Positive")
    else:
        print(r, "-> Negative")

```

This food is delicious and my dog loves it -> Positive
Worst product ever, completely disappointed -> Negative

Inference: The Amazon food review dataset was analyzed using natural language processing techniques. The text data was cleaned and converted into numerical features using TF-IDF vectorization. A Logistic Regression model was trained to classify customer reviews as positive or negative. The model performance was evaluated using accuracy, classification report, and confusion matrix. The results show that the model can effectively identify customer sentiment from review text.